

Tinuola Owolabi

Track-Specific Week 4 Output

AnalystLab Africa — Experience Lab | HealthConnect Clinic Project | Data Analytics Track

1. Problem Statement

HealthConnect Clinic is experiencing a high rate of missed appointments (no-shows) and needs to understand the factors driving this behavior, so data-informed strategies can be developed to reduce no-shows and improve appointment slot efficiency.

2. Dataset Overview

The HealthConnect Appointment Dataset contains 5,000 fictional patient appointment records across 18 columns. It represents a simulated hospital/clinic setting and includes information relating to patient demographics, appointment details, booking information, previous appointment history, previous no-shows, reminder information, distance to the clinic, waiting time, and appointment outcomes. This dataset forms the basis for investigating patterns associated with missed appointments (no-shows).

3. Data Quality Assessment

An initial data quality review was conducted using conditional formatting in Excel to visually scan for missing values and inconsistencies.

- Missing values: Two columns — Distance_to_Clinic_KM and Waiting_Time_Minutes — contain blank entries. These will need to be addressed before deeper analysis, either by labeling them as "Unknown," estimating/imputing values, or excluding affected rows depending on how significant the gap is.
- Duplicates: No duplicate rows were identified using conditional formatting checks.
- Overall: Aside from the two columns with missing values, the dataset appears structurally consistent and ready for further exploration.

4. Relevant Business Questions

- Does distance to the clinic affect the likelihood of a no-show?
- Does receiving an appointment reminder reduce the likelihood of a no-show?
- Does long waiting time during a previous visit affect whether a patient attends their next appointment?
- Does a patient's history of previous no-shows predict whether they will miss future appointments?
- Are certain patient demographics (e.g., gender) associated with different no-show rates?

5. Proposed KPIs

No-Show Rate

Linked to: overall problem. Measures the percentage of total appointments missed; serves as the baseline metric for the whole analysis.

Total Appointments

Linked to: all business questions. Provides the scale/context needed to interpret no-show numbers meaningfully.

Average Distance (No-Shows vs. Attended)

Linked to: Question 1. Compares average distance to the clinic between patients who missed appointments and those who attended, to assess whether distance is a driver of no-shows.

Average Waiting Time (No-Shows vs. Attended)

Linked to: Question 3. Compares average waiting time between patients who missed their next appointment and those who attended, to assess whether waiting time affects future attendance.

Total Patients with Previous No-Shows

Linked to: Question 4. Identifies the scale of patients with a history of missed appointments, to assess whether past behavior predicts future no-shows.

6. Initial Analysis Approach

The proposed analysis approach for future stages of the project includes cleaning the dataset (addressing missing values in `Distance_to_Clinic_KM` and `Waiting_Time_Minutes`), then grouping patients by key factors - distance range, reminder status, prior waiting time, and previous no-show history — and comparing no-show rates across these groups to identify which factors show the strongest association with missed appointments. This may later be supported by visualizations and, where appropriate, statistical analysis to confirm significant relationships.

7. Assumptions, Limitations, Risks and Dependencies

Assumptions

- The dataset is assumed to accurately represent typical appointment booking, reminder, and attendance patterns for the purpose of this exercise.
- Reminder and no-show data are assumed to be recorded correctly at the time of each appointment.

Limitations

- The `Distance_to_Clinic_KM` and `Waiting_Time_Minutes` columns contain missing values, which may limit the completeness of analysis involving these factors until addressed.
- The dataset is fictional and anonymized; findings may not fully reflect real-world patient behavior at an actual healthcare facility.

Risks

- Incomplete or missing data (distance and waiting time) could lead to biased or incomplete conclusions if not properly handled in later analysis.
- Relying on a limited set of factors (distance, reminders, waiting time, history) may overlook other real-world drivers of no-shows not captured in this dataset (e.g., cost, transportation access, illness).

Dependencies

- Further analysis depends on cleaning and addressing the missing values identified in the data quality assessment.
- The reliability of KPI calculations in later weeks depends on the dataset remaining consistent and unaltered from its original structure.